

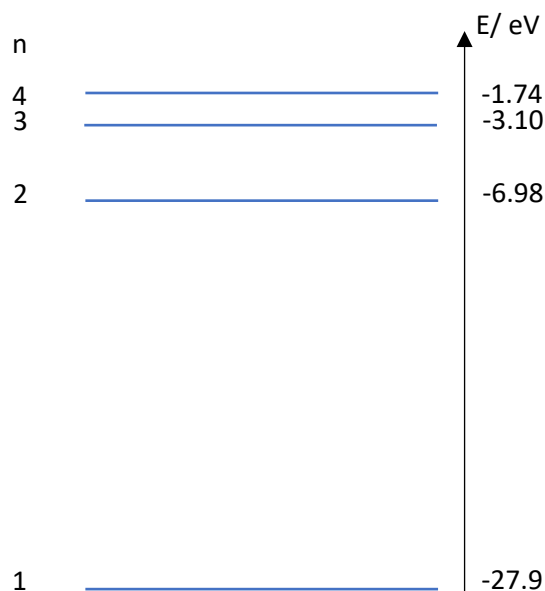
- 6 (a) (i) -When an electron transits from a higher energy level to lower energy level, **energy is released** as electromagnetic wave or in the form of a **photon, of energy hf** . The difference in energy between 2 levels, ΔE is equal to the energy of the photon/ or **$\Delta E = hf$** .
 -since energy level is discrete, the **difference in energy is also discrete**. Hence, only **certain discrete frequencies exist**, hence line spectra B1 obtained.

- (ii) Energy has to be supplied to the electron to bring it to infinity/ to remove the electron, where PE is zero without a change in KE B1
 OR
 The electron is **bound** to the nucleus. Total energy of the system/atom is negative. Hence work has to be done to remove the electron from the atom.

(ii)

Using $E_n = -\frac{27.9}{n^2}$

n	E_n / eV
1	-27.9
2	-6.98
3	-3.10
4	-1.74



All E values correct – B1

Relative spacing between energy levels correct – B1

Diagram must be fully labelled with n and E values – B1

- (iii) Using $\Delta E = E_n - (-27.9) = 27.9 - \frac{27.9}{n^2}$, where ΔE represents the remaining energy of the colliding electron.

n	$\Delta E / \text{eV}$
$1 \rightarrow 2$	20.92
$1 \rightarrow 3$	24.80
$1 \rightarrow 4$	26.16
$1 \rightarrow 5$	26.78

Calculation of ΔE for $1 \rightarrow 4$ **and** $1 \rightarrow 5$

Electron does not have enough energy to excite to $n = 5$

M1

Hence highest energy level reach is $n = 4$

A1

(iv) Using

$$\Delta E = \frac{hc}{\lambda} \quad \Rightarrow \quad \lambda = \frac{hc}{\Delta E}$$

Shortest wavelength

$$\lambda = \frac{hc}{\Delta E_{4 \rightarrow 1}} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{26.16 \times 1.60 \times 10^{-19}} = 47.5 \text{ nm}$$

C1

A1

(b) (i) Maximum energy of electron is equal to energy of photon with shortest wavelength.

$$E_{\max} = \frac{hc}{\lambda_{\min}}$$

[C1]

C1

Accelerating potential

$$V = \frac{hc}{e\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.60 \times 10^{-19} \times 2.00 \times 10^{-10}} = 6220 \text{ V}$$

A1

Note: $\lambda_{\min}$ must be read to $\frac{1}{2}$ **small square**, otherwise minus 1 mark

(ii) -Bombarding high energy electrons **knock out** the inner shell electrons of the target atoms. Hence, electrons transit from higher orbital shells to the vacant inner shells

B1

-These transitions result in the emission of (X-ray) photons whose energies are given by the difference in the energy levels $= \frac{hc}{\lambda}$, resulting in sharp peaks at **specific** wavelengths of $4.00 \times 10^{-10} \text{ m}$, $6.60 \times 10^{-10} \text{ m}$ and $9.95 \times 10^{-10} \text{ m}$.

B1